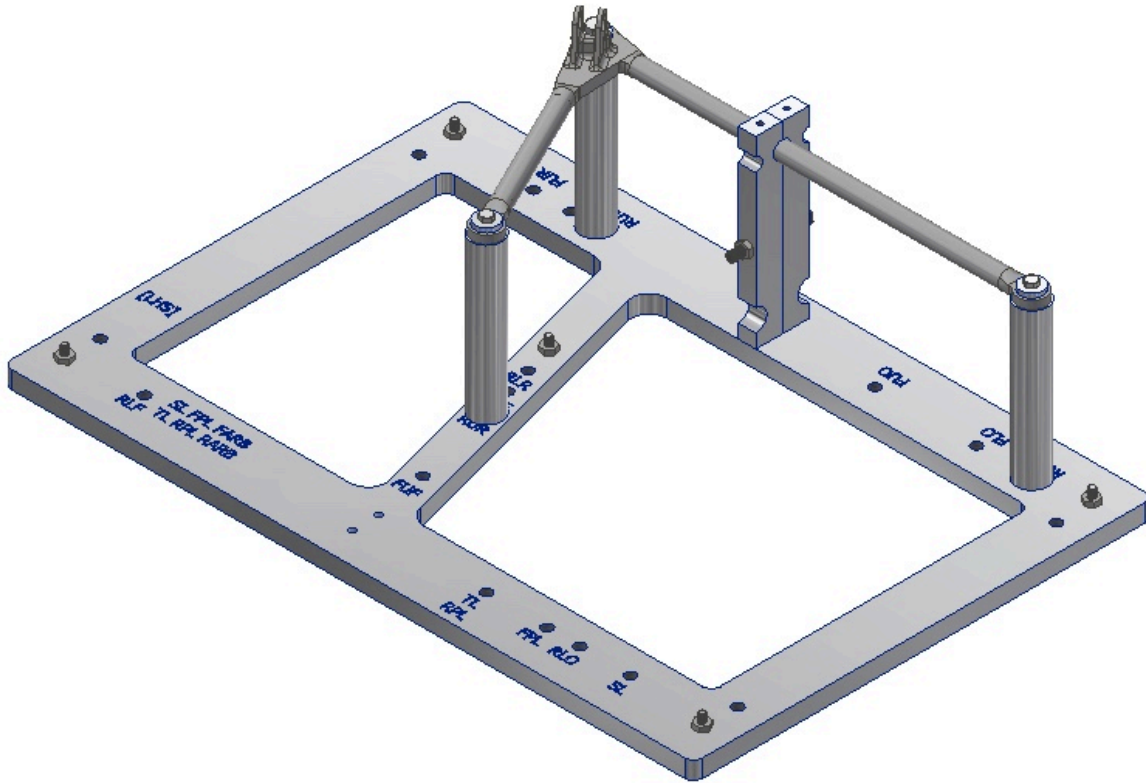


Links Jig Plate
ARG26 Spring Technical Report
May 17, 2026
Davis Rattanavijai



Subteam: Suspension
Subteam Lead: Sujan Munver
Team: Chassis
Technical Team Lead: Amanda Backholm and William Baumohl

1. Introduction

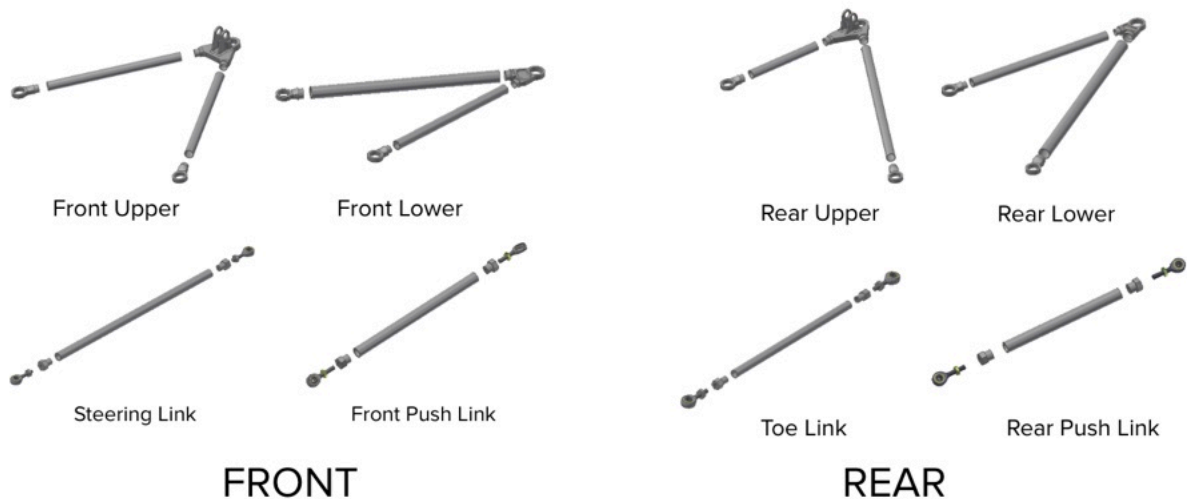
The jig plate is an off-car fixture used to weld each suspension link's components (tubes, inserts, adapters) together into each link's designed geometry. The suspension system consists of 10 unique link types, each designed to vehicle kinematic and structural requirements. The primary goal of the jig plate is to ensure accurate positioning and alignment of all suspension components during welding. Its secondary contribution is enabling a more efficient and repeatable in-house manufacturing process. Because it is an off-car component, it is not governed by FSAE rules.

2. Technical Overview

Steps to create a jig plate.

1. Identify the Links Supported by the Jig Plate

Determine which suspension links the jig plate will locate and constrain. This may also require coordination with the anti-roll bar designer. For example, the ARG26 jig plate supported 8 of the 10 suspension links, which excludes the two anti-roll bar drop links.



2. Define Geometric Constraints

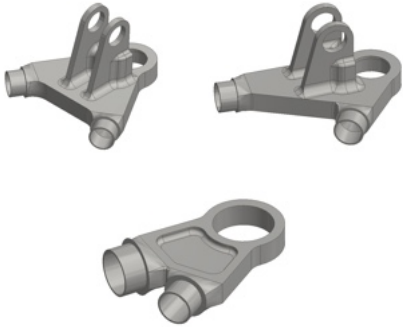
Obtain all required geometric constraints for each link and its associated components (see table below for ARG26). Enter these geometric constraints as CAD parameters as link dimensions can be subject to change. ARG26 jig plate parameters can be found below

sdrive\Reports\2026 Car\Spring
 Technical_Suspension_Jigplate\SU26_A0801_Parameters_(Excel Baseplate Parameters).

Link	Specifications												
Front Upper 	<table> <tr> <th>Link Subpart</th><th>Value</th></tr> <tr> <td>Front Upper Front Length</td><td>9.530 in</td></tr> <tr> <td>Front Upper Rear Length</td><td>9.638 in</td></tr> <tr> <td>Front Upper Angle</td><td>63.821 deg</td></tr> <tr> <td>Front Upper Front Tube</td><td>0.500 in</td></tr> <tr> <td>Front Upper Rear Tube</td><td>0.500 in</td></tr> </table>	Link Subpart	Value	Front Upper Front Length	9.530 in	Front Upper Rear Length	9.638 in	Front Upper Angle	63.821 deg	Front Upper Front Tube	0.500 in	Front Upper Rear Tube	0.500 in
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Front Lower 	<table> <tr> <th>Link Subpart</th><th>Value</th></tr> <tr> <td>Front Lower Front Length</td><td>9.548 in</td></tr> <tr> <td>Front Lower Rear Length</td><td>11.460 in</td></tr> <tr> <td>Front Lower Angle</td><td>33.604 in</td></tr> <tr> <td>Front Lower Front Tube</td><td>0.500 in</td></tr> <tr> <td>Front Lower Rear Tube</td><td>0.625 in</td></tr> </table>	Link Subpart	Value	Front Lower Front Length	9.548 in	Front Lower Rear Length	11.460 in	Front Lower Angle	33.604 in	Front Lower Front Tube	0.500 in	Front Lower Rear Tube	0.625 in
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Rear Upper 	<table> <tr> <th>Link Subpart</th><th>Value</th></tr> <tr> <td>Rear Upper Front Length</td><td>12.290 in</td></tr> <tr> <td>Rear Upper Rear Length</td><td>6.793 in</td></tr> <tr> <td>Rear Upper Angle</td><td>63.816 in</td></tr> <tr> <td>Rear Upper Front Tube</td><td>0.500 in</td></tr> <tr> <td>Rear Upper Rear Tube</td><td>0.500 in</td></tr> </table>	Link Subpart	Value	Rear Upper Front Length	12.290 in	Rear Upper Rear Length	6.793 in	Rear Upper Angle	63.816 in	Rear Upper Front Tube	0.500 in	Rear Upper Rear Tube	0.500 in
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	<table border="1"> <tr> <td>Rear Lower Front Length</td><td>12.283 in</td></tr> <tr> <td>Rear Lower Rear Length</td><td>9.583 in</td></tr> <tr> <td>Rear Lower Angle</td><td>38.843 deg</td></tr> <tr> <td>Rear Lower Front Tube</td><td>0.625 in</td></tr> <tr> <td>Rear Lower Rear Tube</td><td>0.625 in</td></tr> </table>	Rear Lower Front Length	12.283 in	Rear Lower Rear Length	9.583 in	Rear Lower Angle	38.843 deg	Rear Lower Front Tube	0.625 in	Rear Lower Rear Tube	0.625 in
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Steering Link 	<table border="1"> <tr> <th>Link Subpart</th><th>Value</th></tr> <tr> <td>Length</td><td>13.713 in</td></tr> <tr> <td>Tube Diameter</td><td>0.500 in</td></tr> <tr> <td>Rod End Height</td><td>0.312 in</td></tr> <tr> <td>Rod End Inner Diameter</td><td>0.196 in (AN3)</td></tr> </table>	Link Subpart	Value	Length	13.713 in	Tube Diameter	0.500 in	Rod End Height	0.312 in	Rod End Inner Diameter	0.196 in (AN3)
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Rod End Height	0.312 in										
Rod End Inner Diameter	0.196 in (AN3)										
Toe Link 	<table border="1"> <tr> <th>Link Subpart</th><th>Value</th></tr> <tr> <td>Length</td><td>9.650 in</td></tr> <tr> <td>Tube Diameter</td><td>0.375 in</td></tr> <tr> <td>Rod End Height</td><td>0.312 in</td></tr> <tr> <td>Rod End Inner Diameter</td><td>0.196 in (AN3)</td></tr> </table>	Link Subpart	Value	Length	9.650 in	Tube Diameter	0.375 in	Rod End Height	0.312 in	Rod End Inner Diameter	0.196 in (AN3)
Link Subpart	Value										
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Front Push Link 	<table border="1"> <tr> <th>Link Subpart</th><th>Value</th></tr> <tr> <td>Length</td><td>11.364 in</td></tr> <tr> <td>Tube Diameter</td><td>0.625 in</td></tr> <tr> <td>Rod End Height</td><td>0.375 in</td></tr> <tr> <td>Rod End Inner Diameter</td><td>0.250 in (AN4)</td></tr> </table>	Link Subpart	Value	Length	11.364 in	Tube Diameter	0.625 in	Rod End Height	0.375 in	Rod End Inner Diameter	0.250 in (AN4)
Link Subpart	Value										
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Rear Push Link	<table border="1"> <tr> <th>Link Subpart</th><th>Value</th></tr> </table>	Link Subpart	Value								
Link Subpart	Value										

	Length	9.400 in
	Tube Diameter	0.625 in
	Rod End Height	0.375 in
	Rod End Inner Diameter	0.250 in (AN4)

Adapter	Specifications
Outboard (Front Upper, Rear Upper, Front/Upper Lower) 	Inner Diameter = 0.656 in Adapter Height = 0.250 in
Inboard 	Inner Diameter = 0.562 in Adapter Height = 0.219 in

Parameter	Value	Units	Comment	Classification				
FUF_Length	9.530364264	in	Front Upper Front Length	Front Upper	C O N T R O L A R M S			
FUR_Length	9.638114442	in	Front Upper Rear Length					
FU_Angle	63.821017840	deg	Front Upper Angle					
FUF_Tube	0.500000000	in	Front Upper Front Tube					
FUR_Tube	0.500000000	in	Front Upper Rear Tube					
FLF_Length	9.548376878	in	Front Lower Front Length	Front Lower		C O N T R O L A R M S		
FLR_Length	11.460063307	in	Front Lower Rear Length					
FL_Angle	33.604333900	deg	Front Lower Angle					
FLF_Tube	0.500000000	in	Front Lower Front Tube					
FLR_Tube	0.625000000	in	Front Lower Rear Tube					
RUF_Length	12.290405079	in	Rear Upper Front Length	Rear Upper			C O N T R O L A R M S	
RUR_Length	6.793476430	in	Rear Upper Rear Length					
RU_Angle	63.816544100	deg	Rear Upper Angle					
RUF_Tube	0.500000000	in	Rear Upper Front Tube					
RUR_Tube	0.500000000	in	Rear Upper Rear Tube					
RLF_Length	12.283226978	in	Rear Lower Front Length	Rear Lower				C O N T R O L A R M S
RLR_Length	9.583002713	in	Rear Lower Rear Length					
RL_Angle	38.842855960	deg	Rear Lower Angle					
RLF_Tube	0.625000000	in	Rear Lower Front Tube					
RLR_Tube	0.625000000	in	Rear Lower Rear Tube					
SL_Length	13.713281883	in	Steering Link Length	Steering Link	S T R A I G H T L I N K S			
SL_Tube	0.500000000	in	Steering Link Tube	Toe Link				
SL_RodEnd_H	0.312000000	in	Steering Link Rod End Height					
TL_Length	9.649581442	in	Toe Link Length					
TL_Tube	0.375000000	in	Toe Link Tube	Push Links				
TL_RodEnd_H	0.312000000	in	Toe Link Rod End Height					
FPL_Length	11.364228966	in	Front Push Link Length					
FPL_Tube	0.625000000	in	Front Push Link Tube	ARB Links				
RPL_Length	9.399986809	in	Rear Push Link Length					
RPL_Tube	0.625000000	in	Rear Push Link Tube					
PL_RodEnd_H	0.375000000	in	Push Link Rod End Height	ARB Links		S T R A I G H T L I N K S		
FARB_Length	2.248000000	in	Front ARB Link Length					
FARB_Tube	0.000000000	in	Front ARB Link Tube					
RARB_Length	2.864000000	in	Rear ARB Link Length					
RARB_Tube	0.000000000	in	Rear ARB Link Tube					
ARB_RodEnd_H	0.000000000	in	ARB Drop Link Rod End Height	Adapter Specs	M I S C			
AD_Tolerance	0.000500000	in	5 thou tolerance. Outsourced part					
Inboard_AD_ID	0.561000000	in	Inner diameter of inboard adapter					
Outboard_AD_ID	0.655000000	in	Inner diameter of inboard adapter					
Inboard_AD_Height	0.219000000	in	Height from bottom of adapter to top					
Outboard_AD_Height	0.250000000	in	Height from bottom of adapter to bottom of flange	AN3/AN4			M I S C	
AN3_Diameter	0.196000000	in	AN3 size for nut					
AN4_Diameter	0.250000000	in	AN4 size for nut					
AN3_Rodend_Height	0.312500000	in	AN3 Rod end height from McMaster					
AN4_Rodend_Height	0.375000000	in	AN4 Rod end height from McMaster					
ControlArm_Height	4.500000000	in	Height at which control arms are being raised	Other		M I S C		
AluminumStock_Diameter	0.875000000	in	Aluminum Stock Diameter					
Stamp_Size	0.313500000	in	Square size of stamp for engraving					
BearingPostThread_Diameter	0.375000000	in	Diameter of Bearing Post					

ARG26 Jig Plate CAD Parameters

3. Establish Jig Plate Architecture

Determine the overall jig plate architecture, including:

- How links and bearing cups/adapters will be constrained and located
- Welding accessibility and sequencing
- Fixture repeatability and usability

For example, the ARG26 jig plate followed a similar architecture to ARG25, while differing significantly from the ARG23 and ARG24 designs.

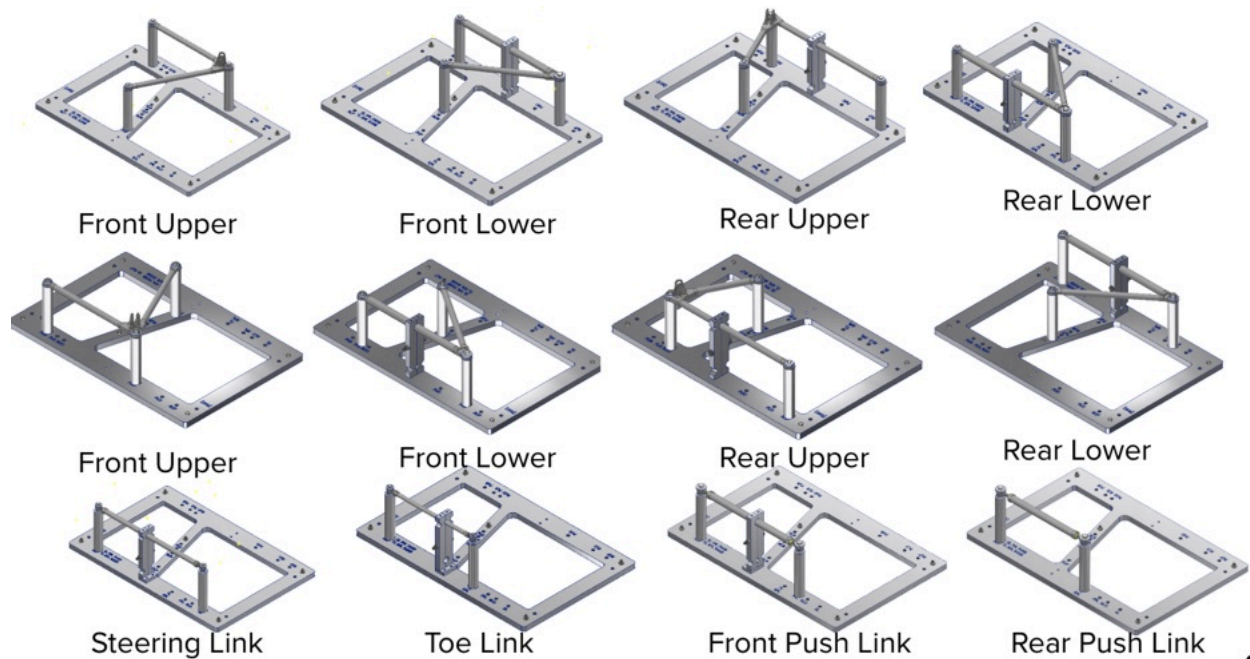
4. Design the Baseplate Layout

5. Design Constraint Features and Hardware

- Bearing posts
- Midpoint posts
- Washers
- Bolts
- Additional locating or clamping features as required

Some feedback to cover are: ease of assembly and operation, welding accessibility and manufacturability.

Assemble all suspension links and jig components in CAD, and using model states is highly recommended. Make sure to use the links CAD. This step is critical for verifying compatibility between the jig plate and all links, identifying interference or tolerance issues, and confirming the intended welding and assembly procedure.



8. Finally, manufacture the jig plate and its components!

3. History of Past Designs

Recent suspension jig plate designs have consistently used precision fixturing with dedicated hole placements for each link configuration. Across all recent iterations, the core principle has remained the same: using a combination of holes, bearing posts, and midpoint posts to accurately suspend and constrain suspension links during welding.

The primary architectural change occurred in ARG24, which introduced adjustable fixturing. However, due to the complexity and time required for setup, the adjustability feature was removed in ARG25 in favor of a simpler fixed-layout approach.

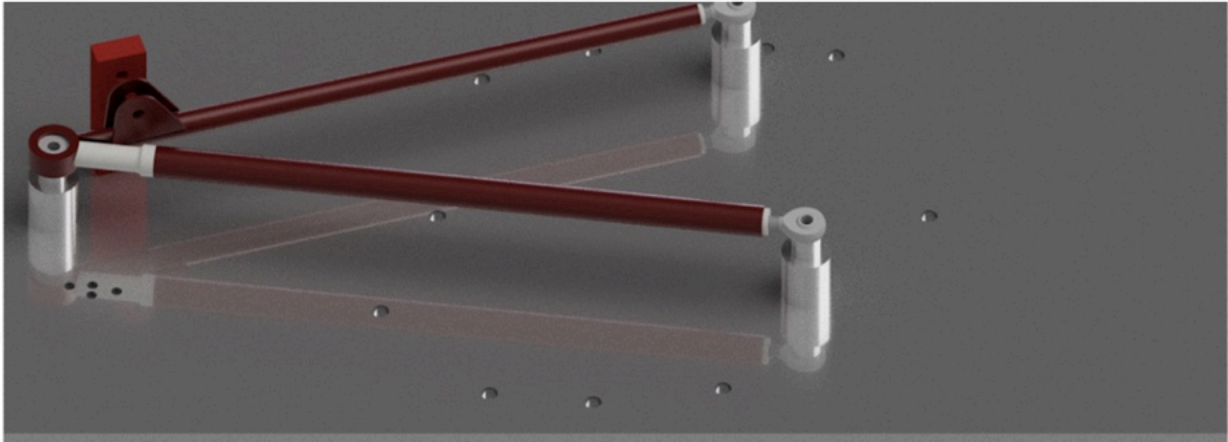
ARG21 - Andrew Kurth

The ARG21 jig plate highlighted several early manufacturing and organizational challenges. Andrew Kurth noted in his suspension manufacturing report [1] that inconsistent post heights and bolt sizes caused setup issues during manufacturing, leading to incorrect fixturing and the need to discard some components.

To address these issues, several improvements were proposed:

- Properly labeled and stamped posts
- Accurate accounting for bearing and rod-end heights to ensure links rested level
- Dedicated ball-joint posts for COM-4T and COM-5T bearings
- Integrated AN4 and AN5 thread configurations to eliminate unnecessary spacers

A key lesson from ARG21 was the importance of maintaining link levelness efficiently. While spacers and additional hardware can compensate for geometric mismatches, minimizing unnecessary components greatly improves setup simplicity, repeatability, and manufacturing efficiency.



ARG21 Jig Plate CAD Render (Source: Andrew Kurth's Report [1])

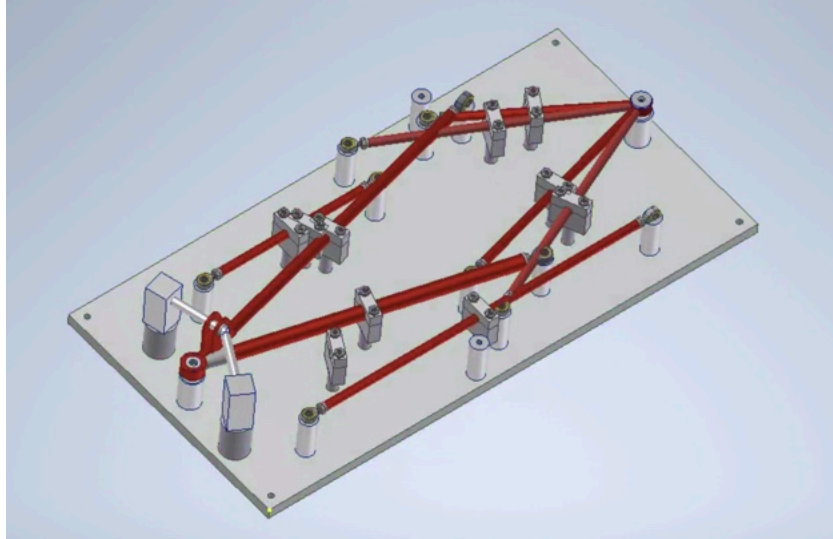
ARG22 - Zach Buttrick and Ben Ehemann

From Zach Buttrick's report [2], the ARG22 jig plate focused heavily on improving organization, documentation, and assembly workflow. First, Rather than maintaining a single complex CAD assembly, separate assemblies were created for each linkage configuration. This simplified drawing generation and reduced confusion during manufacturing and setup.

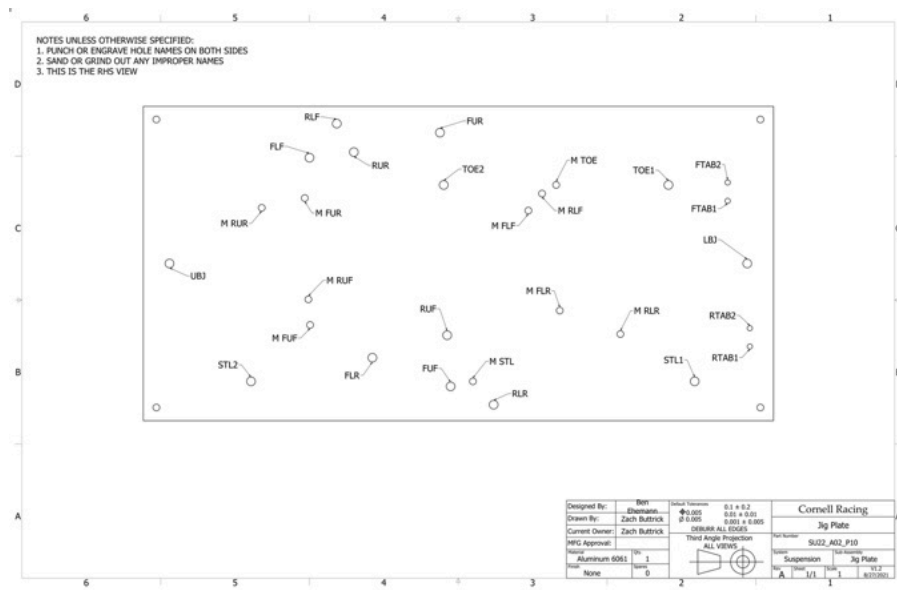
Major improvements included:

- Large-format (11" × 17") jig plate drawings
- Dedicated assembly drawings for each linkage setup
- Comprehensive BOM documentation organized into a single binder
- Engraved labels directly machined into the jig plate
- Additional support holes across the baseplate for improved rigidity
- Organized storage systems for fixture hardware and posts

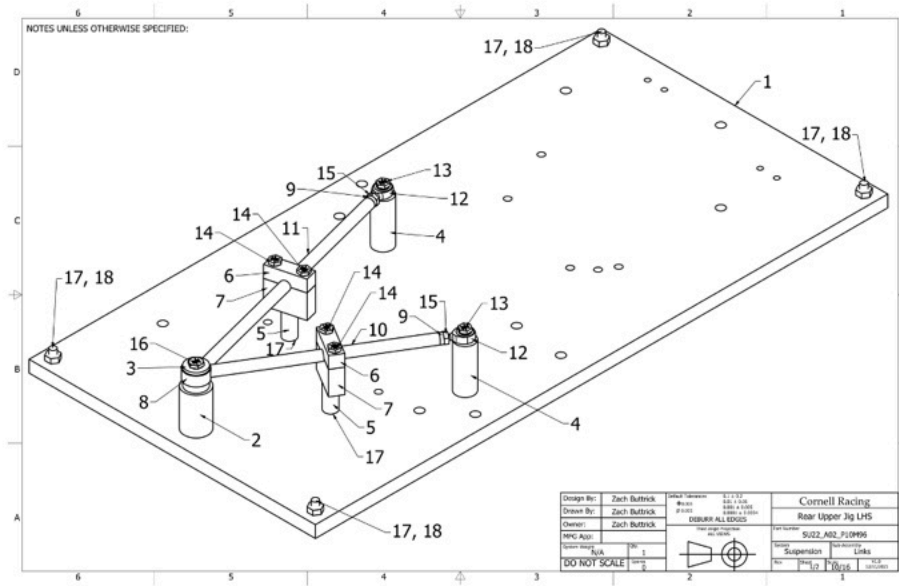
These changes significantly improved repeatability and allowed team members unfamiliar with the jig plate to assemble link setups more efficiently.



ARG22 Jig Plate CAD (Source: Zach Buttrick's Report [2])



ARG22 Jig Plate Drawing with Engravings (Source: Zach Buttrick's Report [2])



ARG22 Jig Plate Assembly Drawing (Source: Zach Buttrick's Report [2])

REAR UPPER LINKS ASSEMBLY LHS

ITEM	PART NUMBER	DESCRIPTION	QUANTITY
1	SU22_A02_P10	Jig Plate	1
2	SU22_A02_P12	Upper B1 Post	1
3	SU22_A02_P18	Upper B1 Top	1
4	SU22_A02_P14	AN3 Post	2
5	SU22_A02_P19	Midpoint Support Bar	2
6	SU22_A02_P20	Midpoint Support Top Uppers	2
7	SU22_A02_P23	Midpoint Support Bottom Uppers	2
8	SU22_A02_P08	Upper Bearing Cup	1
9	SU22_A02_P02	AN3 Insert	2
10	SU22_A020201_P01	RUF Link	1
11	SU22_A020201_P02	RUR Link	1
12	NA	AN-3 Aurora Rod End	2
13	NA	AN-5 Bolt	2
14	NA	AN-7 Bolt	4
15	NA	AN3 Checknut LoPro	2
16	NA	AN4-4 Bolt	1
17	NA	AN4-7 Bolt	6
18	NA	AN4 Checknut	4

Design By: Zach Buttrick
 Drawn By: Zach Buttrick
 Owner: Zach Buttrick
 MFG App: [blank]
 Title: [blank]
 DO NOT SCALE

Check Version: [blank]
 # of Revs: 1
 Date: 1/1/2020
 Drawn By: Zach Buttrick
 Owner: Zach Buttrick
 MFG App: [blank]
 Title: [blank]
 DO NOT SCALE

Cornell Racing
 Rear Upper Jig LHS
 SU22_A02_P10M96
 Suspension Links
 Rev: 1/1/2020

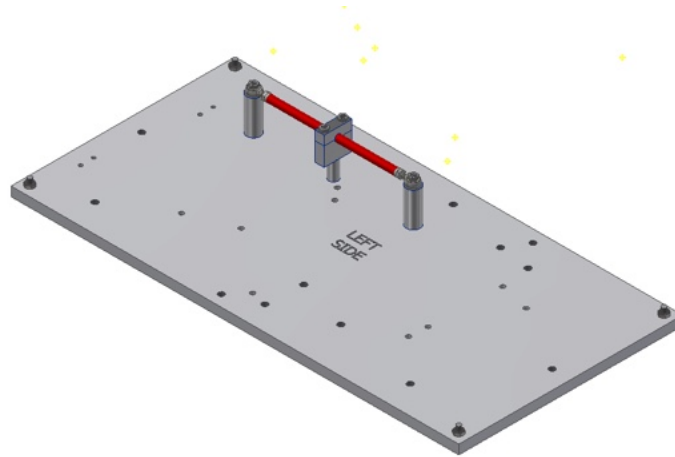
ARG22 Jig Plate Example BOM (Source: Zach Buttrick's Report [2])

ARG23 - Lexi Stoff

Although ARG23 did not introduce major architectural changes from ARG22. One major concern was that links were removed from the jig plate to complete underside welds. Removing the links before cooling allowed thermal distortion to occur off-fixture, reducing dimensional consistency between manufactured parts.

Feedback from ARG23 strongly influenced the development priorities of later jig plate designs, particularly the need for:

- Improved fixturing rigidity
- Better weld accessibility
- The ability to keep links constrained throughout the full welding process



ARG23 Jig Plate

ARG24 - Joaquin Paz

ARG24 represented the largest change in jig plate design and documentation. Joaquin Paz and Amanda Backholm both wrote valuable information about this in their technical reports, referenced [4] and [5] respectively.

Its defining feature was the introduction of adjustable fixturing. Adjustable inboard mounting locations provided multiple degrees of freedom, allowing positional corrections to compensate for monocoque manufacturing deviations. This flexibility also enabled the fabrication of spare links with modified geometries when required.

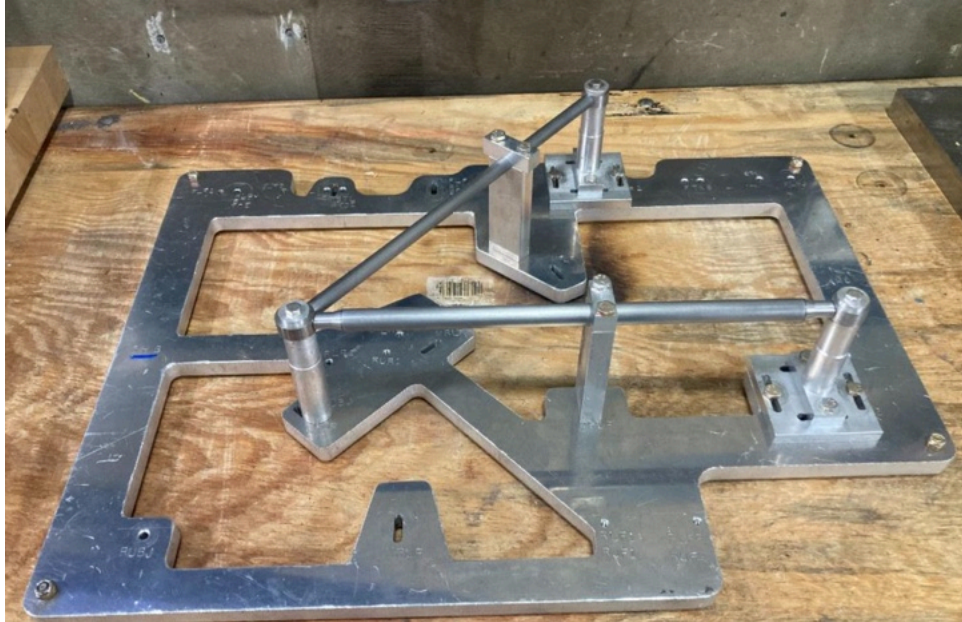
Another major improvement was the ability to keep links fully constrained throughout the entire welding process, including underside welds. This significantly reduced thermal distortion and warping compared to previous years.

Key features included:

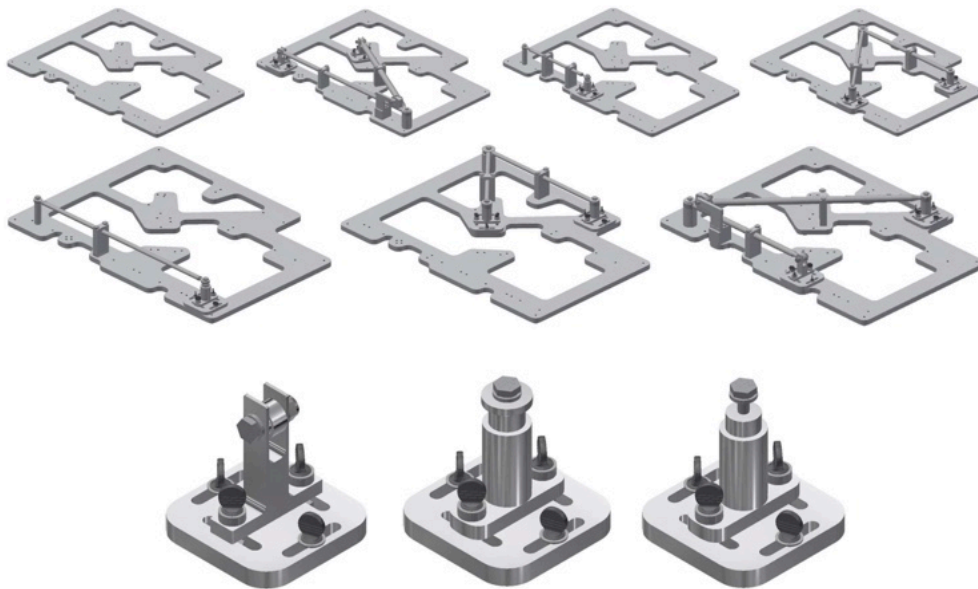
- Adjustable locating systems
- Modular metal posts
- Configurable tab fixturing
- Support for multiple bearing cup orientations and link geometries

Despite these advancements, several issues were identified after manufacturing:

- Tight slot tolerances required additional post-machining
- Insufficient cutouts restricted welder access to some joints
- Plated fasteners released toxic fumes during welding
- Overall setup and adjustment procedures were time-consuming and cumbersome



ARG24 Jig Plate Assembly (Source: Amanda Backholm's Report [4])



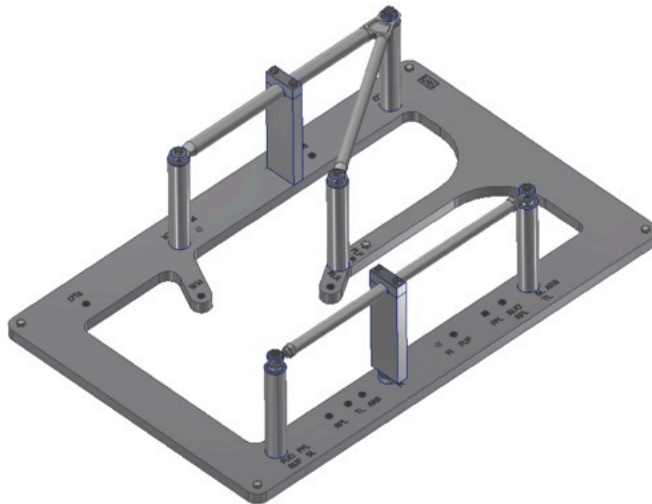
ARG24 Jig Plate CAD Renderings (Source: Joaquin Paz's Report [5])

ARG25 - Sujan Munver

According to Sujan Munver's reports [6] and [7], ARG25 simplified the architecture introduced in ARG24 by removing adjustability and transitioning back to a fixed-layout jig plate. This was made possible through the links team's adoption of adapters, which reduced the need for adjustable fixturing. The ARG25 design emphasized manufacturability, weld accessibility, and setup efficiency and demonstrated fairly well accuracy with 7 of 8 links fitting.

Improvements included:

- Maximized cutouts for significantly improved weld access
- Outboard points positioned near all four corners of the plate
- Die-threaded 3/8" bearing posts threaded directly into the baseplate
- Elimination of the need to flip the plate during setup
- Posts raised approximately 1 inch for better welding clearance
- Non-plated stainless steel fasteners replacing plated hardware
- Engraved labels machined during the same CNC operation as hole drilling
- Corner elevation bolts to lift the jig plate off the welding table



ARG25 Jig Plate (Source: Sujan Munver's Report [6])

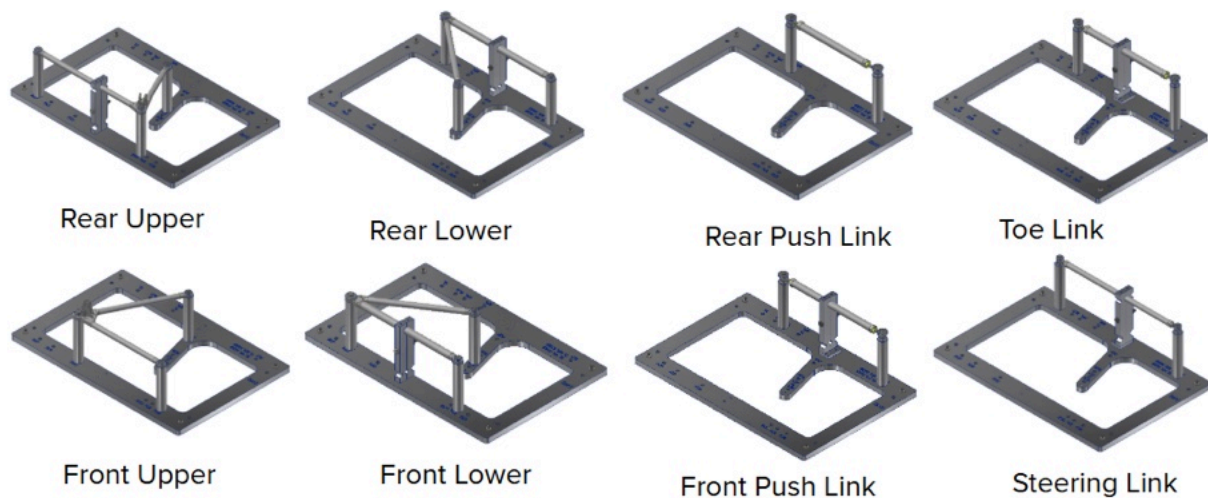
4. ARG26 Design

The goal of the ARG26 jig plate was to optimize and refine the ARG25 design while maintaining a conservative and reliable design philosophy. Similar to the overall suspension philosophy for ARG26, the jig plate intentionally avoided drastic architectural changes in favor of improving proven concepts and ensuring reliable manufacturing within the team's timeline constraints.

The primary design goals for ARG26 were:

- Retain the successful design choices from ARG25
- Resolve usability issues and manufacturing annoyances identified in ARG25
- Increase welding clearance and accessibility
- Reduce overall component count

The final ARG26 jig plate supports all 8 primary suspension links. Although the vehicle contains 10 total links, the anti-roll bar drop links were excluded because their approximately 2-inch length made jig plate fixturing unnecessary.



Baseplate Design

The ARG26 baseplate underwent several design iterations focused on balancing welding accessibility, structural rigidity, and manufacturing simplicity. Throughout the design process, all geometries were constrained directly using the CAD assemblies of the suspension links. Since ARG25's material selection, threading strategy, and general fixturing hardware had already proven successful, the primary optimization effort focused on baseplate geometry and cutout placement.

from Caroline Moskal, the primary suspension link welder, to verify ease of welding.

Key observations from the design iterations included insufficient welding clearance, concerns regarding structural rigidity of a non-rectangle baseplate, and unsupported cantilevered support. The final design combined the strengths of previous iterations to maximize weld accessibility while maintaining sufficient stiffness and a stable rectangular geometry.

Bearing Posts and Washers

The ARG26 bearing posts remained largely unchanged from ARG25.

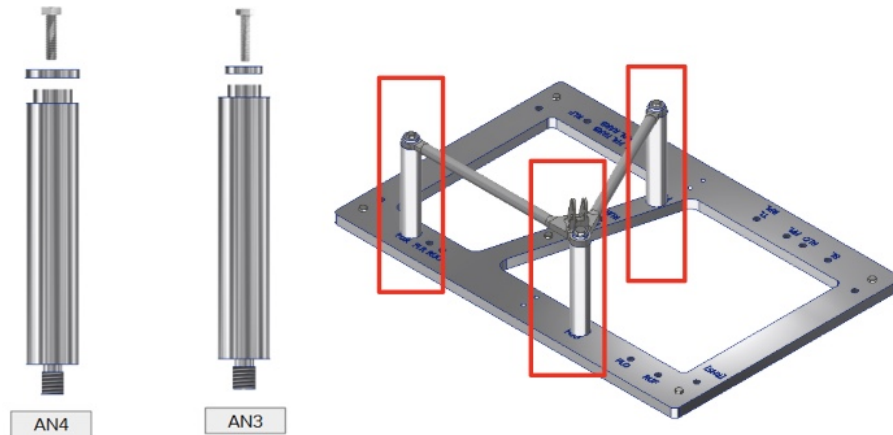
The tall post design was retained because increased post height improves welding accessibility around the joints. However, taller posts also increase the potential for geometric tolerance stack-up and misalignment. Previous designers and alumni feedback suggested that both the post height and post radius contribute to fixturing precision. Although no formal tolerance analysis was conducted, the ARG25 post geometry had previously demonstrated acceptable performance, so the same dimensions were retained for ARG26.

A significant discussion point during Alumni Design Review was the use of threaded posts for precision locating. Threaded mounting provides a simple and effective locating strategy, but practical manufacturing experience revealed some concerns:

- One baseplate thread stripped during use
- Thread damage could critically affect jig accuracy and manufacturing timelines
- Tolerance in threads can introduce misalignment of bearing posts

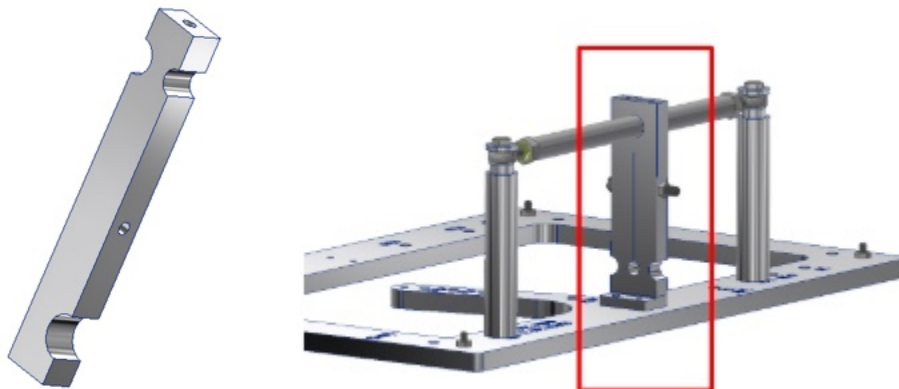
While the threaded design was ultimately sufficient, future iterations should investigate alternative mounting methods that reduce the risk of thread failure without significantly increasing setup complexity.

Another manufacturing consideration involved the use of custom non-plated washers. These washers were intentionally selected to avoid toxic fumes generated by plated hardware during welding. Future designs should investigate commercially available stainless steel washer alternatives to simplify hardware sourcing and reduce dependency on custom components.



Midpoint Support Post

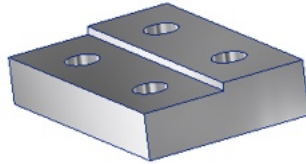
ARG26 introduced a redesigned midpoint support post intended to reduce component count and simplify setup. Previous jig plates used separate midpoint supports for each tube diameter. ARG26 instead consolidated these into a single universal midpoint support capable of accommodating all three tube diameters. However, post-manufacturing feedback revealed several usability drawbacks. Because the supports were installed laterally rather than vertically, assembling links onto the fixture required additional maneuvering and alignment effort. Positioning the tube simultaneously onto the midpoint support and bearing posts often required significant adjustment during setup. For ARG27, it may be beneficial to revisit the previous vertically assembled midpoint support architecture while still exploring opportunities to combine multiple tube diameters into fewer support variants.



ARG26 Midpoint Post (Left) and Example Use (Right)

Midpoint Support Shims

Midpoint support shims are used to elevate midpoint supports to the appropriate height for the steering, toe, and push links. ARG26 consolidated the shim system into a single two-height component rather than using separate shims for each elevation to reduce component count and simplify set up. Although functional, one geometric issue became apparent during use. The width of each shim section matched the midpoint support exactly, which allowed the residual machining radius between the two height steps to introduce slight misalignment. While the resulting error was small, future designs should remove this interference.



ARG26 Midpoint Shim Design

Overall ARG26 feedback

Overall, the ARG26 jig plate successfully achieved its primary objectives and supported the manufacturing timeline without major setbacks. The most significant improvement compared to ARG25 was the substantial increase in welding clearance and accessibility.

However, several opportunities for future improvement remain:

- Reducing dependence on threaded locating features
- Improving midpoint support usability
- Further simplifying assembly procedures
- Reducing sensitivity to manufacturing tolerances

It's important to remember that the jig plate is fundamentally a manufacturing tool, and every design decision should consider the complete assembly workflow step-by-step. The design process not only physically for the part, but procedurally for links welding should be considered.

5. Manufacturing and Assembly

For ARG26, the baseplate was manufactured by SendCutSend, while the bearing posts, midpoint supports, shims, and washers were machined in-house. Nuts and bolts were purchased from McMaster.

Using SendCutSend significantly reduced manufacturing labor for the baseplate, but some post-processing was still required. Several threaded holes needed to be rethreaded, and additional deburring/chamfering was necessary despite selecting the deburring option during ordering. The in-house machined components were manufactured successfully.

The ARG26 timeline worked well overall. The baseplate was ordered before Thanksgiving and arrived around December 1st, 2025, several weeks before winter break. This schedule provided enough time for post-processing operations such as deburring, stamping, and rethreading, while also leaving margin to reorder the plate if major issues had been discovered. Future teams should aim to maintain a similar timeline. Similarly, the in-house machined components were completed around when the baseplate arrived.

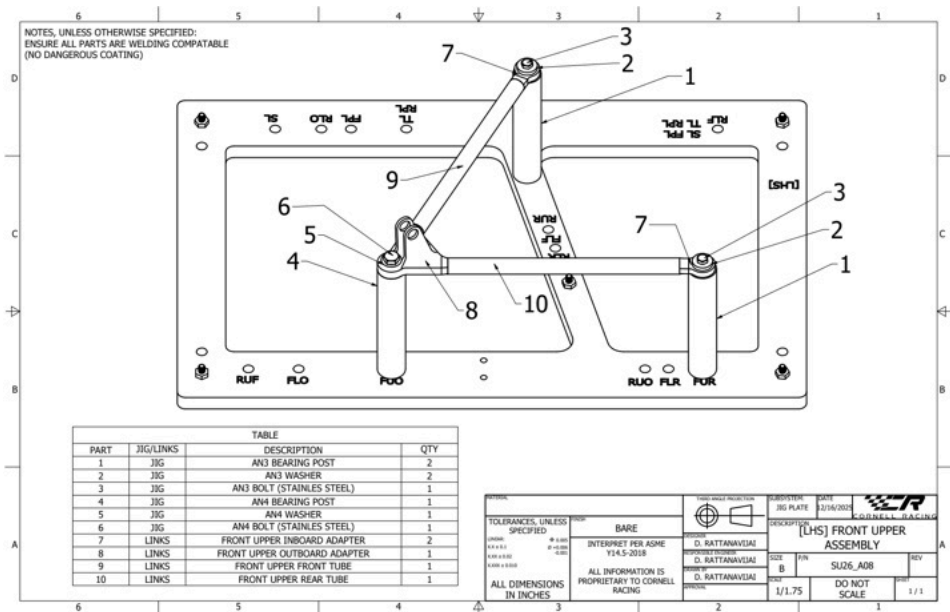
Looking ahead to ARG27, the Emerson machine shop is expected to receive new equipment including a waterjet and laser cutter. This creates the possibility of manufacturing future baseplates entirely in-house. Future teams should evaluate how these new resources could improve manufacturing flexibility, reduce lead times, and further streamline the overall suspension manufacturing schedule.



Once all jig plate components were completed, a binder containing jig plate drawings, setup information, BOMs, and other manufacturing references was assembled to support the suspension links manufacturing process. One particularly useful addition was a one-page reference sheet listing tube sizes and cut lengths for each link.

Although the links designer is ultimately responsible for suspension link manufacturing, the jig plate designer often finishes earlier in the schedule. Because of this, it is valuable for

the jig plate designer to help streamline the overall manufacturing process through improved organization, documentation, and setup resources.



Link	Dimension	Value	Unit	Description
FRONT UPPER	FUF_Length	9.530364264	in	Front Upper Front Length
	FUF_Tube_Length	7.074	in	Front Upper Front Tube Length
	FUF_Tube_OD	0.5	in	Front Upper Front Tube OD
	FUF_Tube_thickness	0.028	in	Front Upper Front Tube Thickness
	FUR_Length	9.638114442	in	Front Upper Rear Length
	FUR_Tube_Length	7.182	in	Front Upper Rear Tube Length
	FUR_Tube_OD	0.5	in	Front Upper Rear Tube OD
	FUR_Tube_thickness	0.028	in	Front Upper Rear Tube Thickness
	FU_Angle	63.82101784	deg	Front Upper Angle
FRONT LOWER	FLF_Length	9.548376878	in	Front Lower Front Length
	FLF_Tube_Length	7.545	in	Front Lower Front Tube Length
	FLF_Tube_OD	0.5	in	Front Lower Front Tube OD
	FLF_Tube_thickness	0.049	in	Front Lower Front Tube Thickness
	FLR_Length	11.46006331	in	Front Lower Rear Length
	FLR_Tube_Length	9.457	in	Front Lower Rear Tube Length
	FLR_Tube_OD	0.625	in	Front Lower Rear Tube OD
	FLR_Tube_thickness	0.049	in	Front Lower Rear Tube Thickness
	FL_Angle	33.6043339	deg	Front Lower Angle
REAR UPPER	RUF_Length	12.29040508	in	Rear Upper Front Length
	RUF_Tube_Length	10.425	in	Rear Upper Front Tube Length
	RUF_Tube_OD	0.5	in	Rear Upper Front Tube OD
	RUF_Tube_thickness	0.028	in	Rear Upper Front Tube Thickness
	RUR_Length	6.79347643	in	Rear Upper Rear Length
	RUR_Tube_Length	4.083	in	Rear Upper Rear Tube Length
	RUR_Tube_OD	0.5	in	Rear Upper Rear Tube OD
	RUR_Tube_thickness	0.028	in	Rear Upper Rear Tube Thickness
	RU_Angle	63.8165441	deg	Rear Upper Angle
REAR LOWER	RLF_Length	12.28322698	in	Rear Lower Front Length
	RLF_Tube_Length	10.57	in	Rear Lower Front Tube Length
	RLF_Tube_OD	0.625	in	Rear Lower Front Tube OD
	RLF_Tube_thickness	0.028	in	Rear Lower Front Tube Thickness
	RLR_Length	9.583002713	in	Rear Lower Rear Length
	RLR_Tube_Length	7.847	in	Rear Lower Rear Tube Length
	RLR_Tube_OD	0.625	in	Rear Lower Rear Tube OD
	RLR_Tube_thickness	0.028	in	Rear Lower Rear Tube Thickness
	RL_Angle	38.84285596	deg	Rear Lower Angle
STEERING	SL_Length	13.71328188	in	Steering Link Length
	SL_Tube_Length	11.337	in	Steering Link Tube Length
	SL_Tube_OD	0.5	in	Steering Link Tube OD
	SL_Tube_thickness	0.035	in	Steering Link Tube Thickness
TOE	TL_Length	9.649581442	in	Toe Link Length
	TL_Tube_Length	7.454	in	Toe Link Tube Length
	TL_Tube_OD	0.375	in	Toe Link Tube OD
	TL_Tube_thickness	0.035	in	Toe Link Tube Thickness
FRONT PUSH	FPL_Length	11.36422897	in	Front Push Link Length
	FPL_Tube_Length	8.858	in	Front Push Link Tube Length
	FPL_Tube_OD	0.625	in	Front Push Link Tube OD
	FPL_Tube_thickness	0.049	in	Front Push Link Tube Thickness
REAR PUSH	RPL_Length	9.399986809	in	Rear Push Link Length
	RPL_Tube_Length	6.978	in	Rear Push Link Tube Length
	RPL_Tube_OD	0.625	in	Rear Push Link Tube OD
	RPL_Tube_thickness	0.028	in	Rear Push Link Tube Thickness

ARG26 Tube Cutting Reference Sheet

6. Test Results

No tests were run or planned.

7. References

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